

POWER BI WEEK-8

REPORT

Introduction

Week 8 focuses on creating an **interactive Student Performance Analytics Dashboard using Power BI**. The prepared dataset from Week 7 is used to analyze students' overall academic performance, subject-wise scores, parental education, test preparation, lunch type, and grade distribution. KPI cards, charts, slicers, and a drill-through **Academic Profile** page are created to provide detailed and interactive analysis. The dashboard is designed to answer key questions about academic performance and identify meaningful patterns in the student data. Finally, the dashboard is validated and evidence-based insights are derived from the analysis.

NAME: Durga Devi S

DEPARTMENT: B.Sc Data Science and Artificial Intelligence

YEAR: II Year

ROLL NO: E25DA018

Figure 1: Opening the prepared Student Performance dataset and verifying the data model and DAX measures for Week 8.

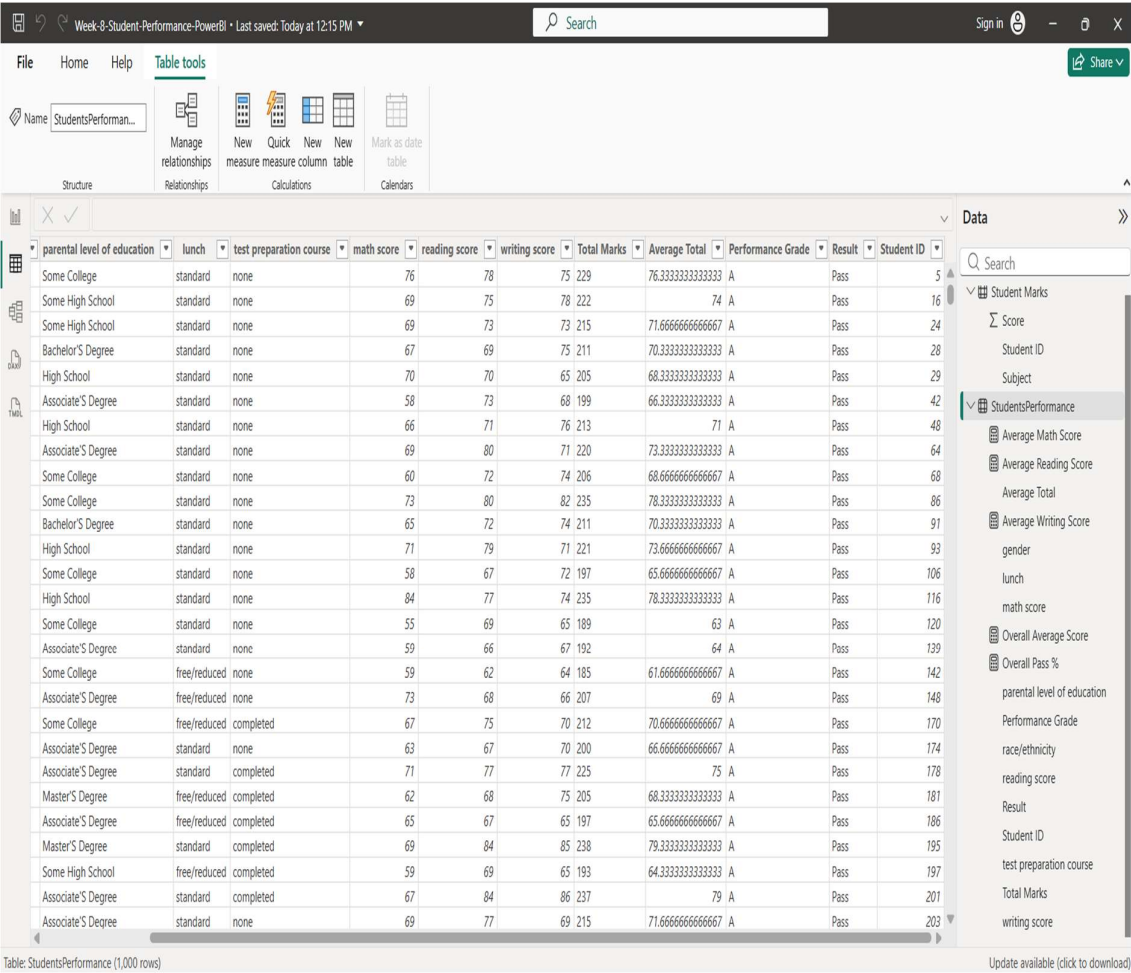


Figure 2: Displaying the Overall Average Score using a KPI card.

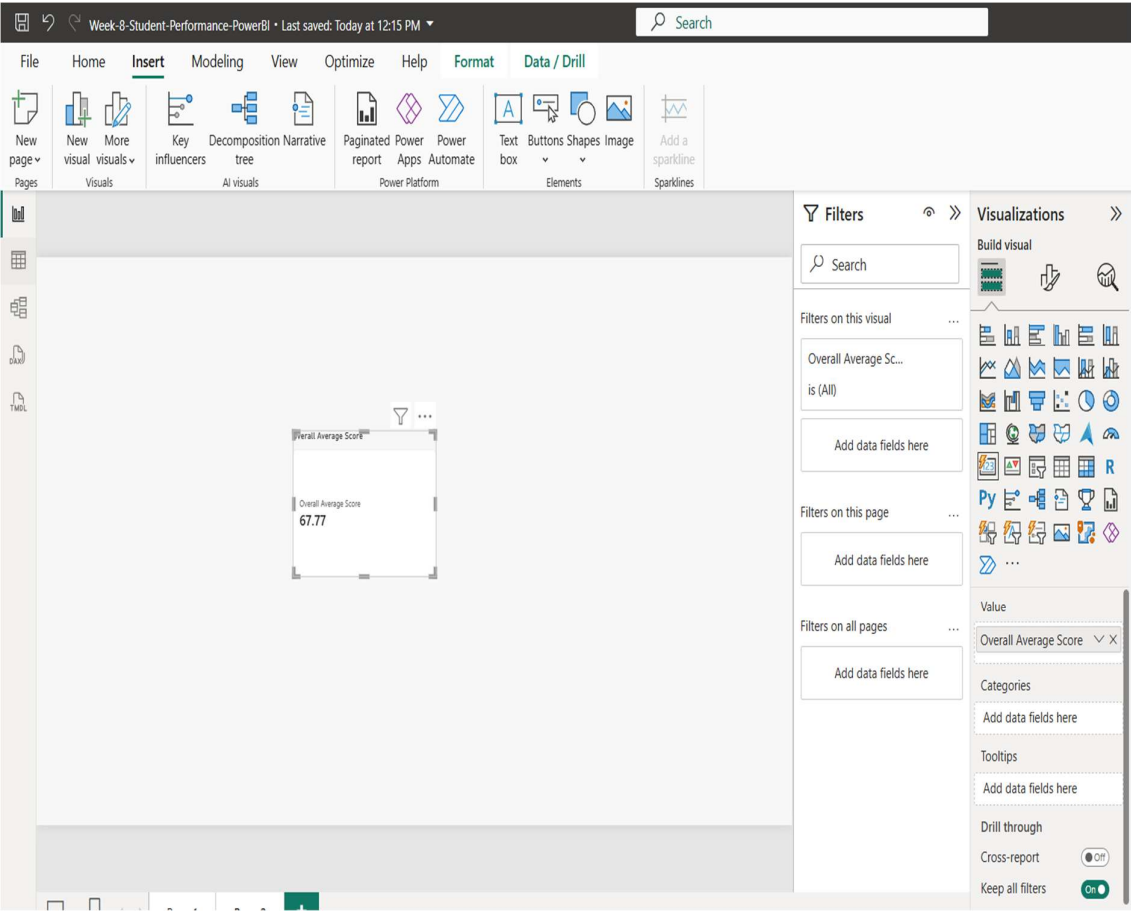


Figure 3: Displaying the Average Math Score using a KPI card.

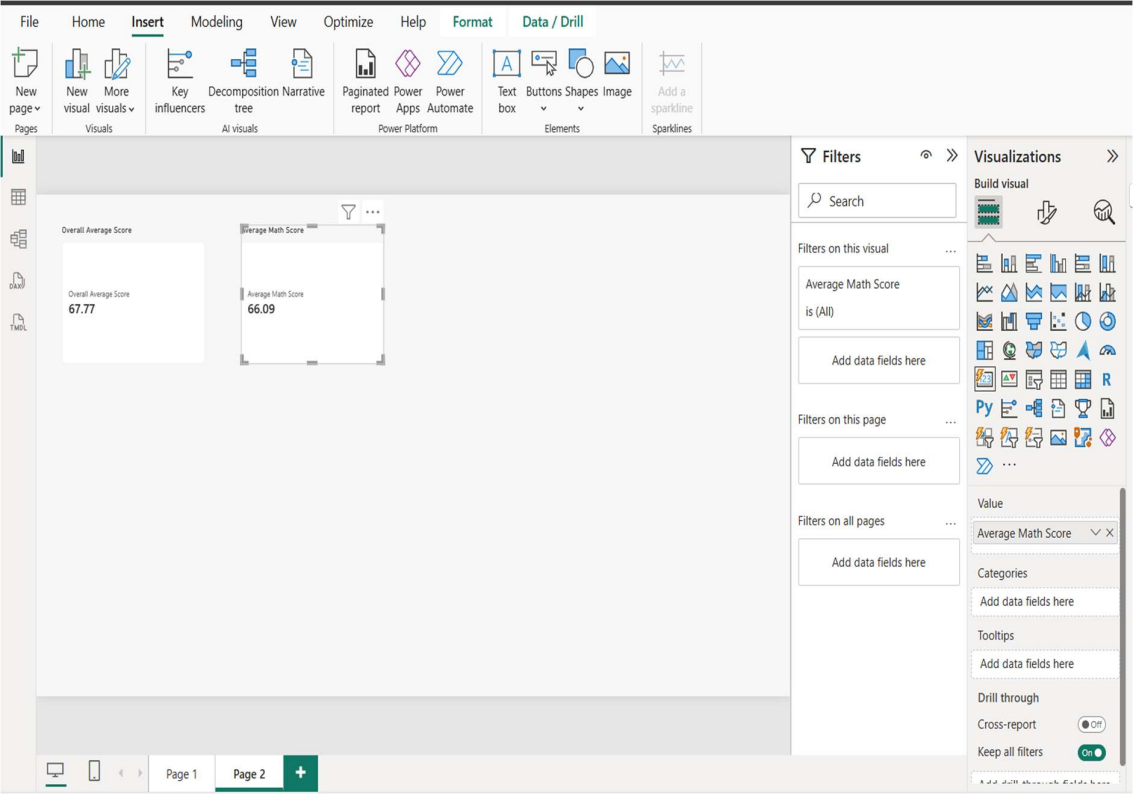


Figure 4: Displaying the Average Reading Score using a KPI card.

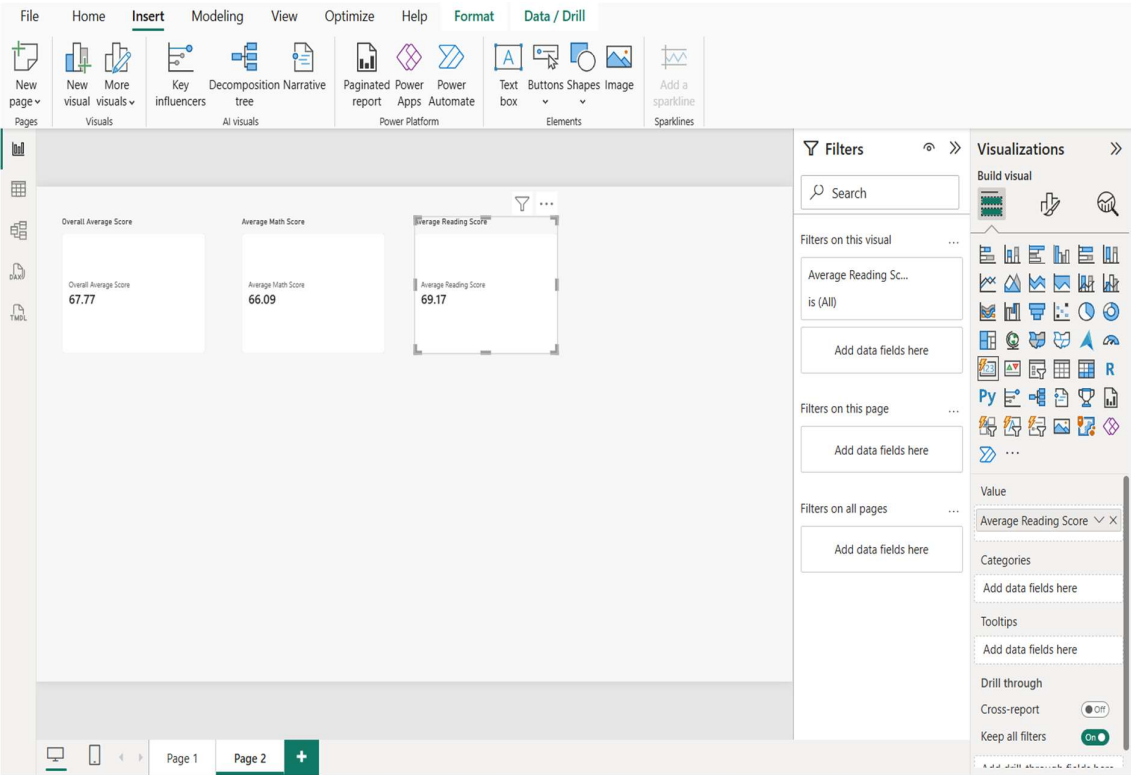


Figure 5: Displaying the Average Writing Score using a KPI card.

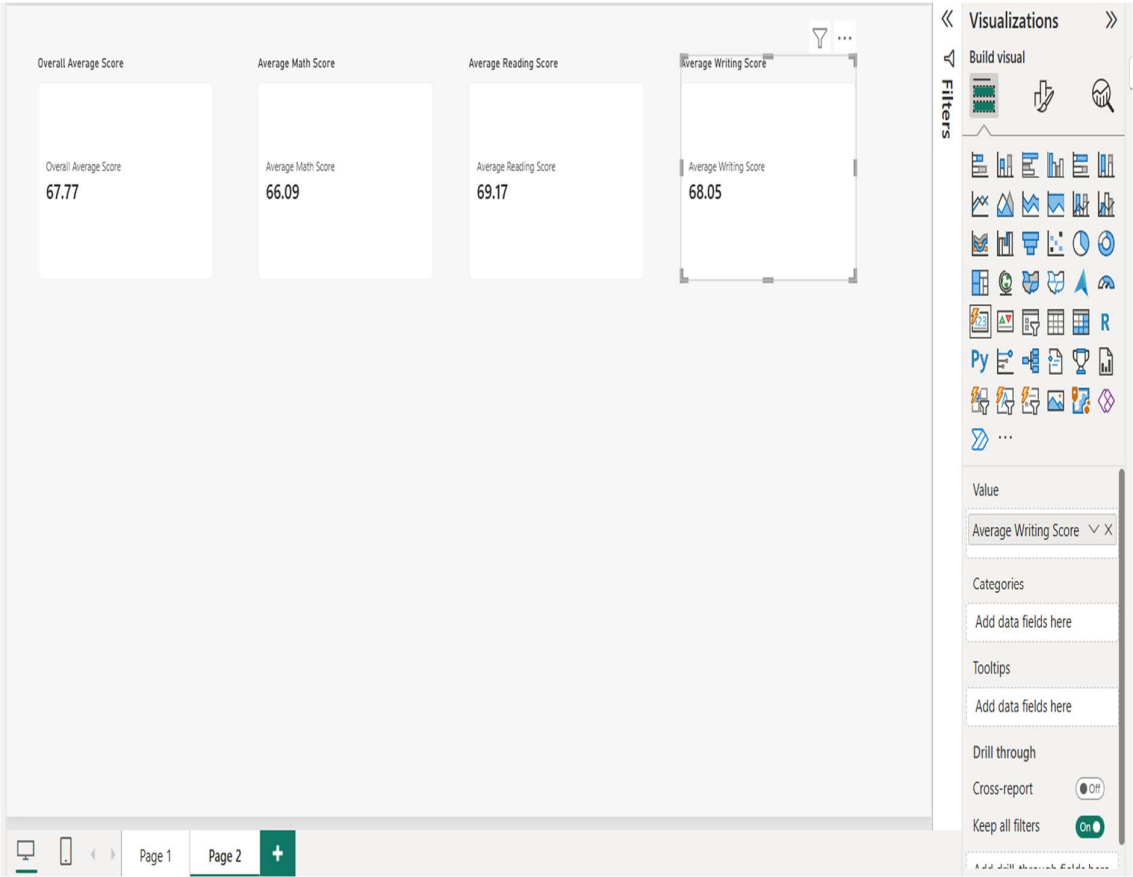


Figure 6: Displaying the Overall Pass Percentage using a KPI card.

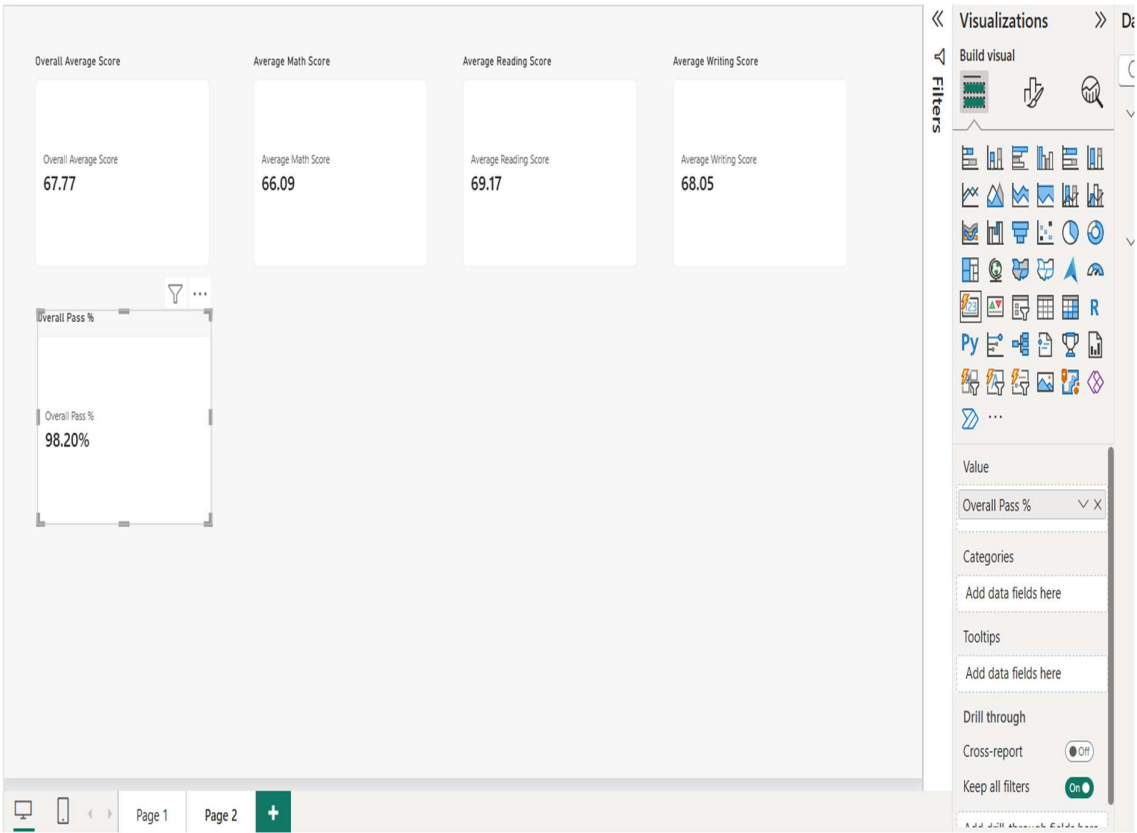


Figure 7: Comparing average student performance across Math, Reading, and Writing subjects.

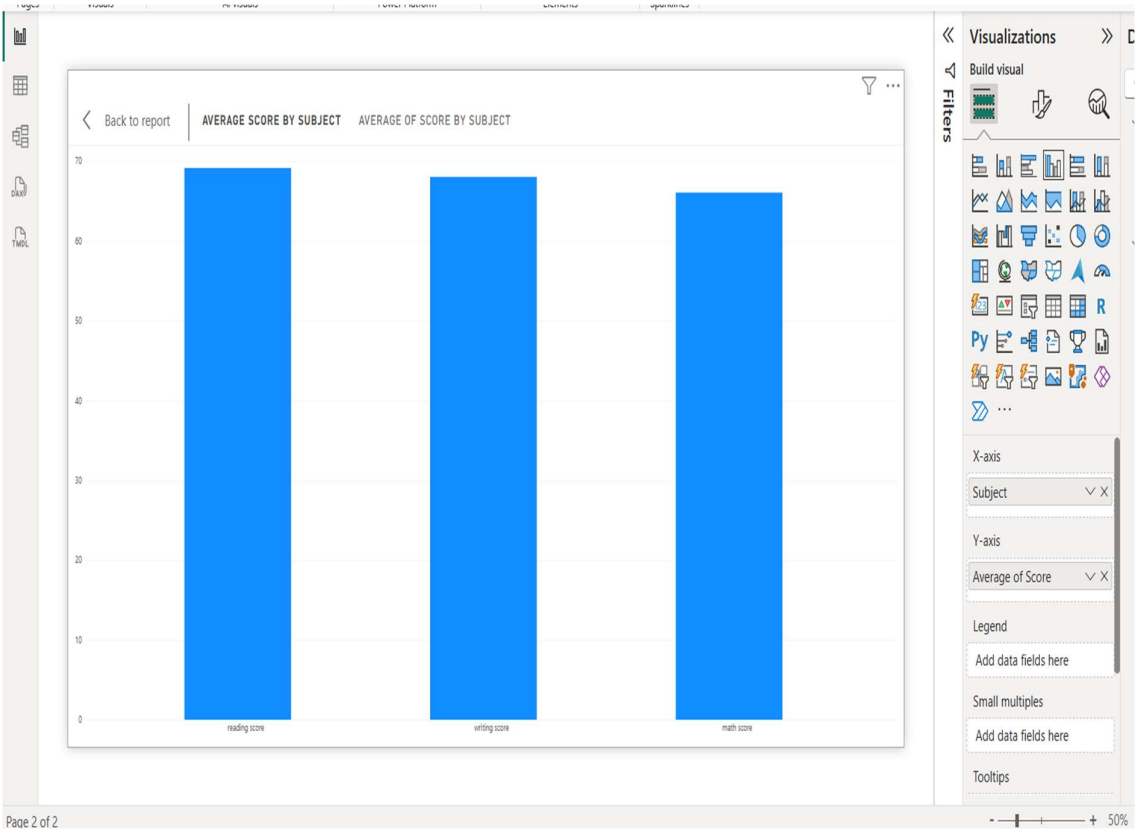


Figure 8: Comparing average student scores across parental education levels.



Figure 9: Comparing average student performance based on test preparation course completion.

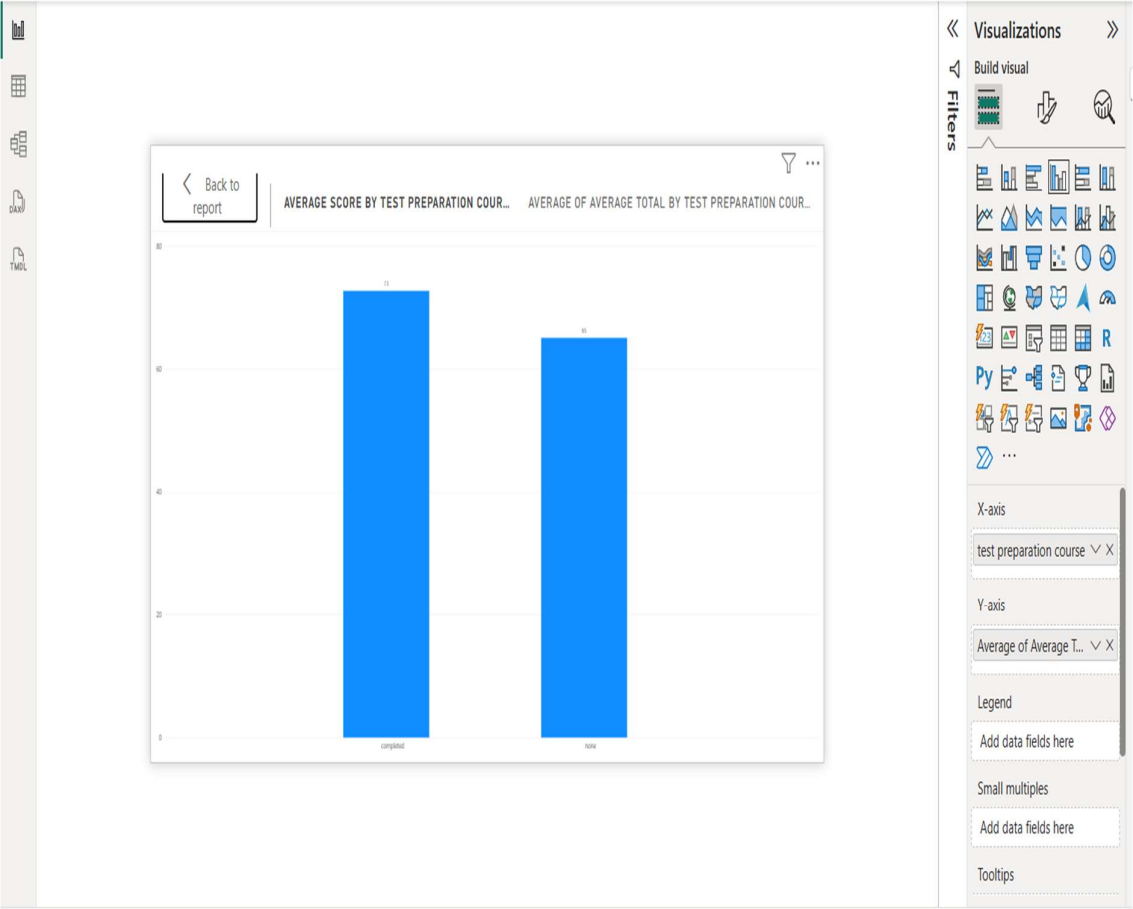


Figure 10: Comparing average student performance based on lunch type.

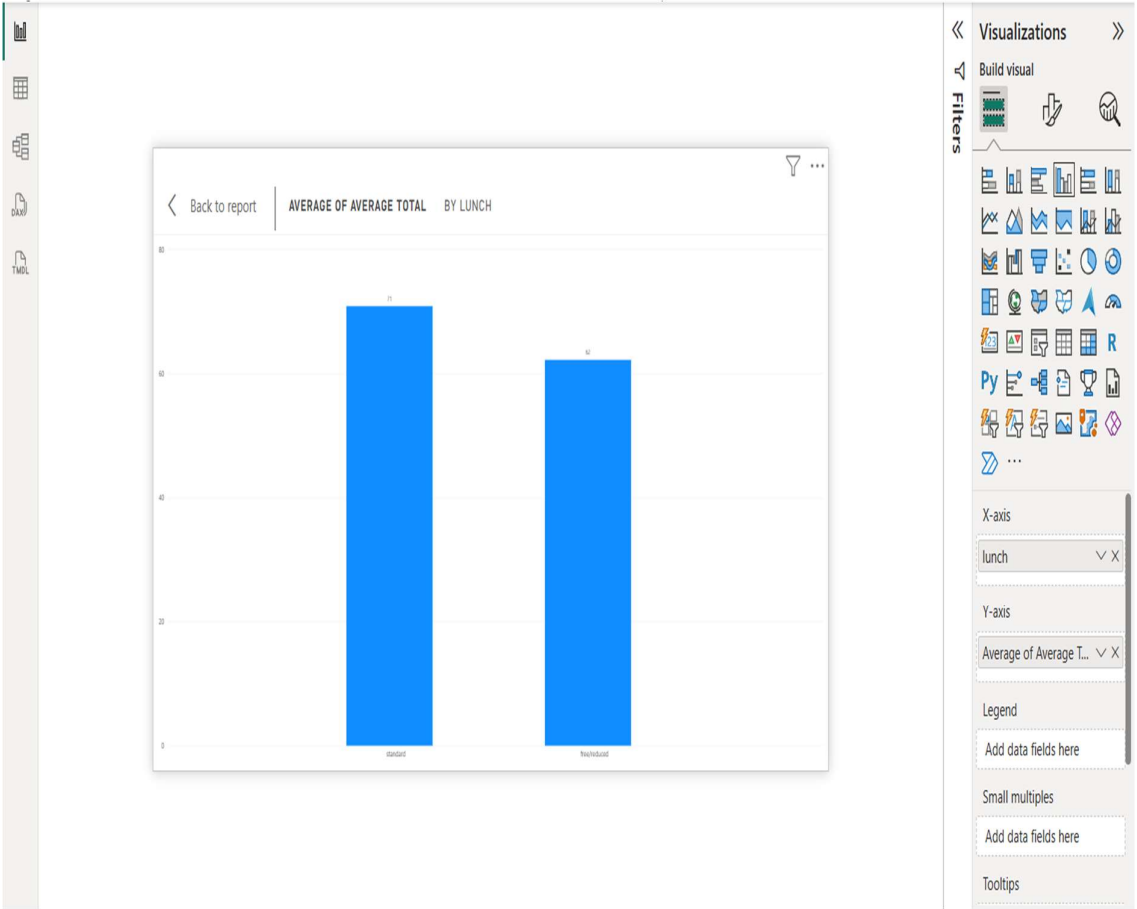


Figure 11: Showing the distribution of students across different Performance Grade categories.

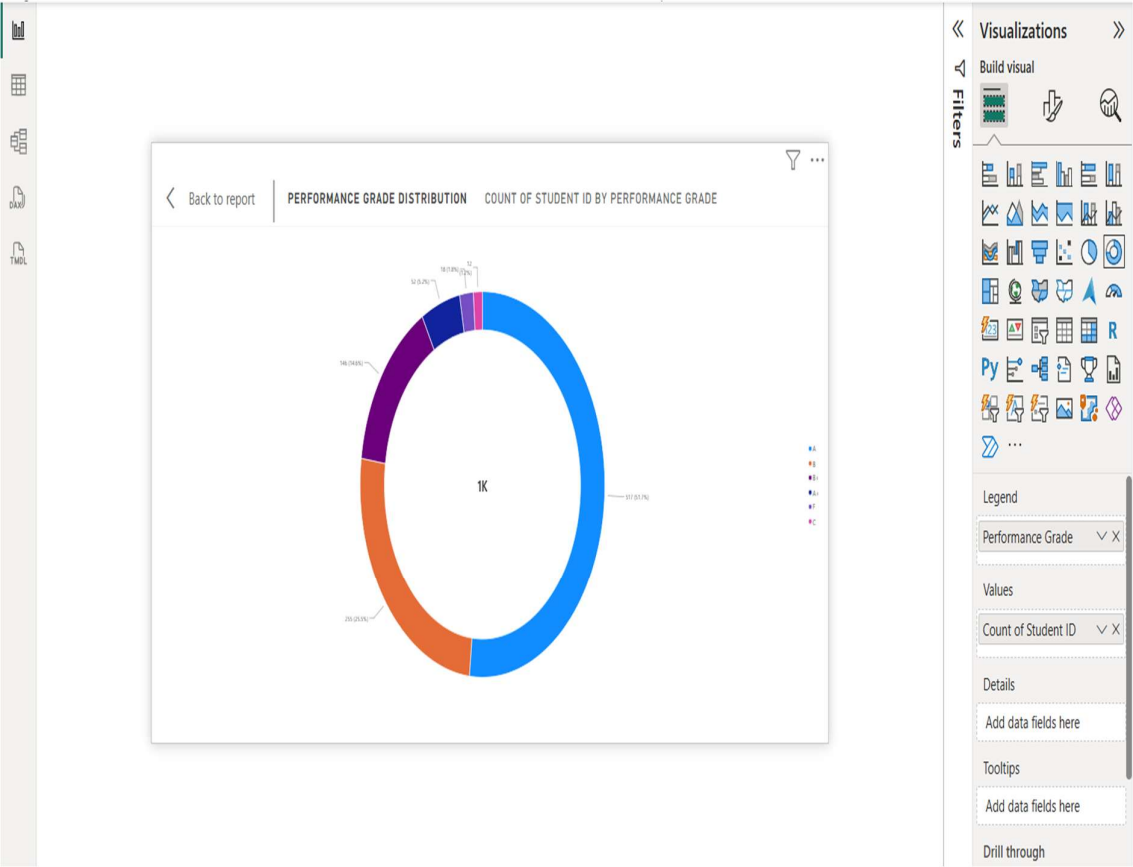


Figure 12: Adding a Gender slicer for interactive student performance analysis.

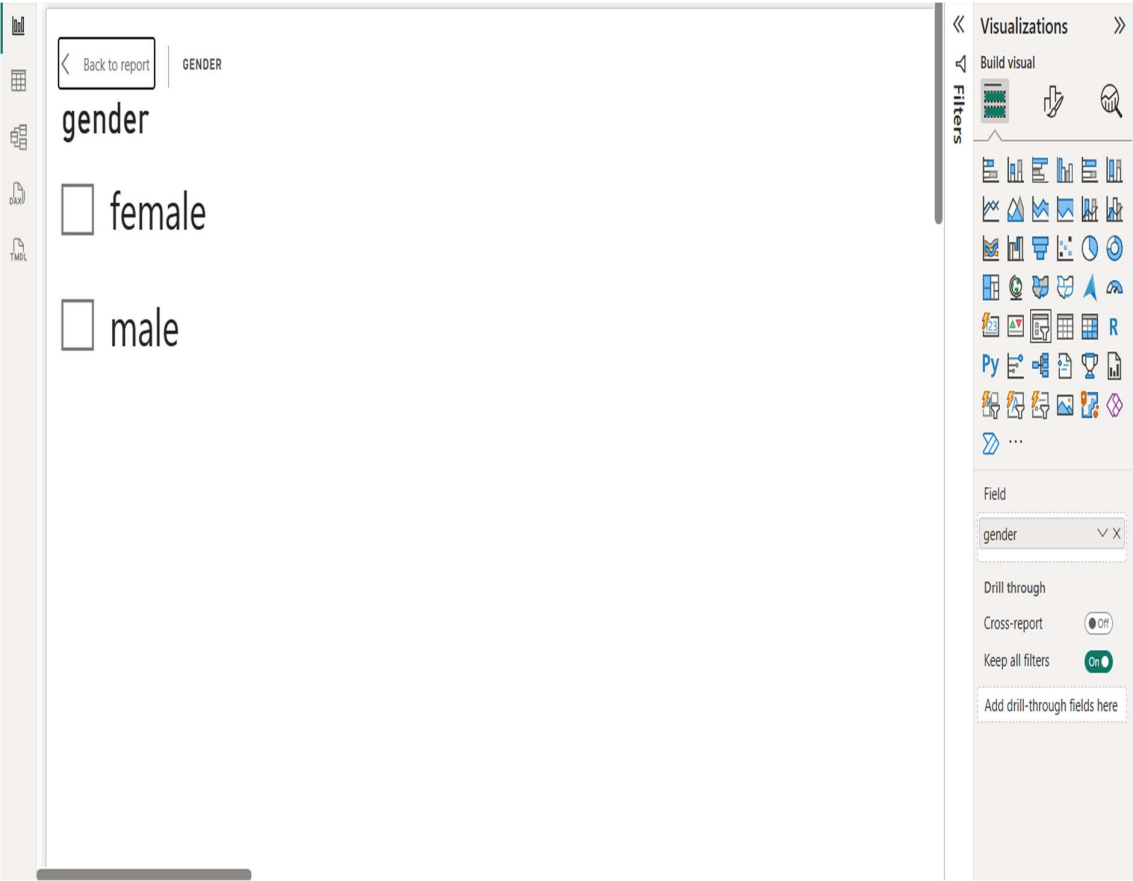


Figure 13: Adding a Race/Ethnicity slicer for interactive student performance analysis.

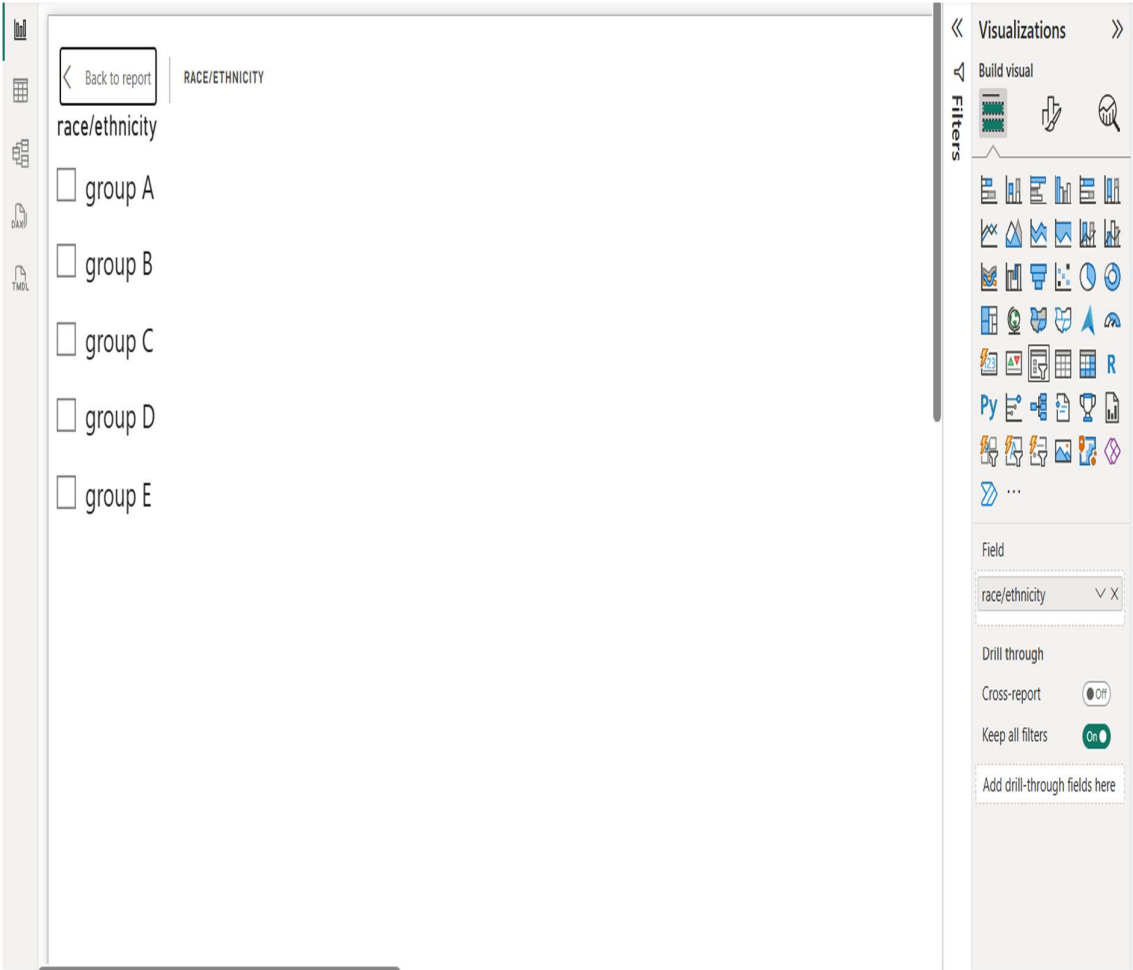


Figure 14: Adding a Parental Education slicer for interactive student performance analysis.

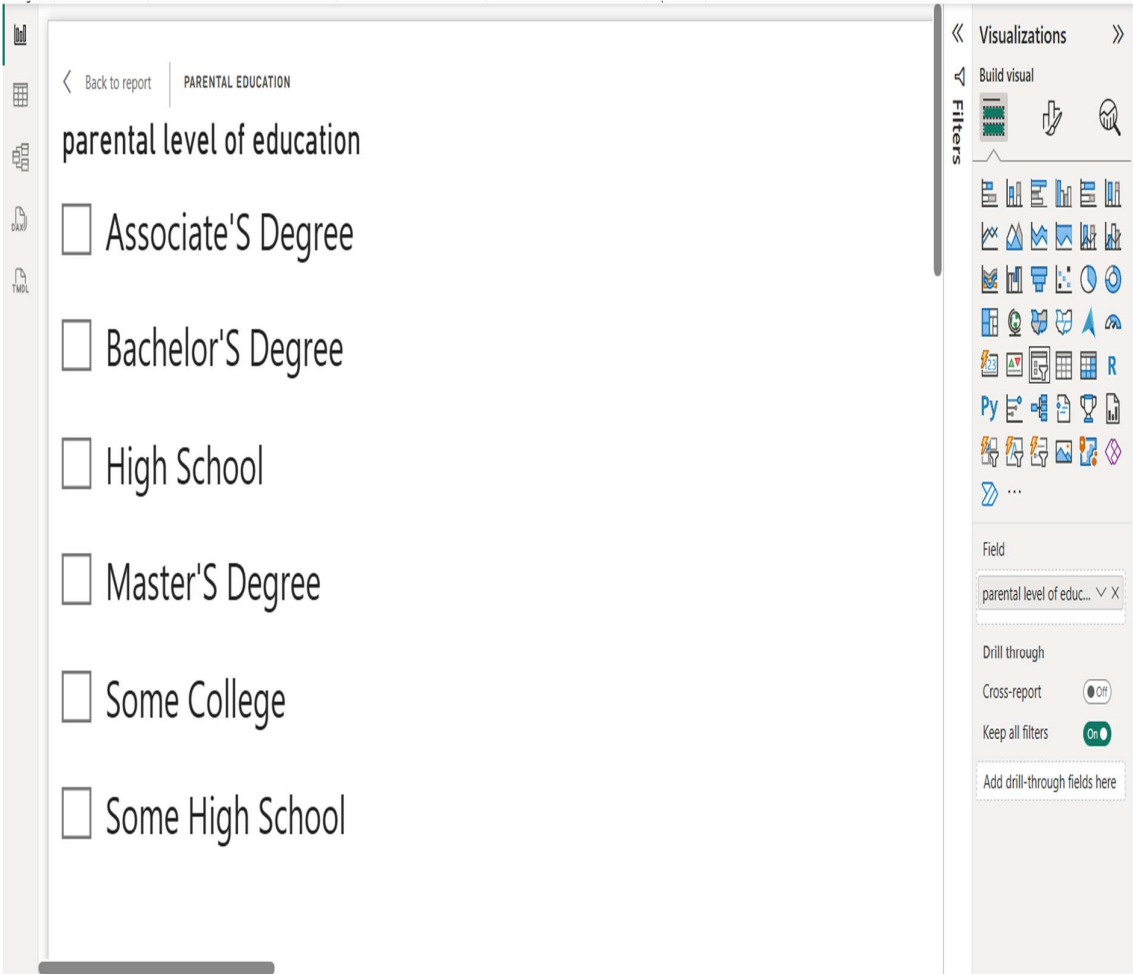


Figure 15: Adding a Lunch Type slicer for interactive student performance analysis.

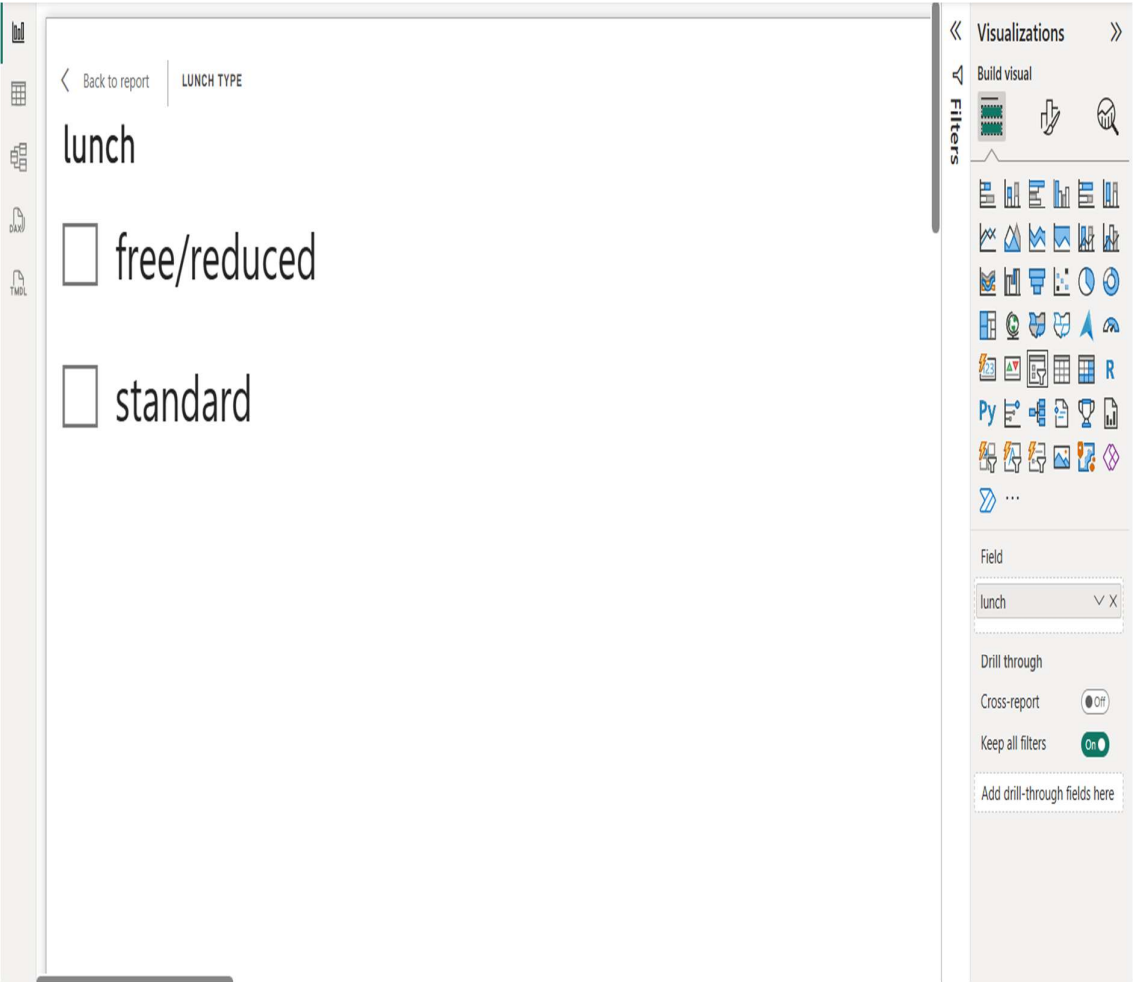


Figure 16: Adding a Test Preparation Course slicer for interactive student performance analysis.

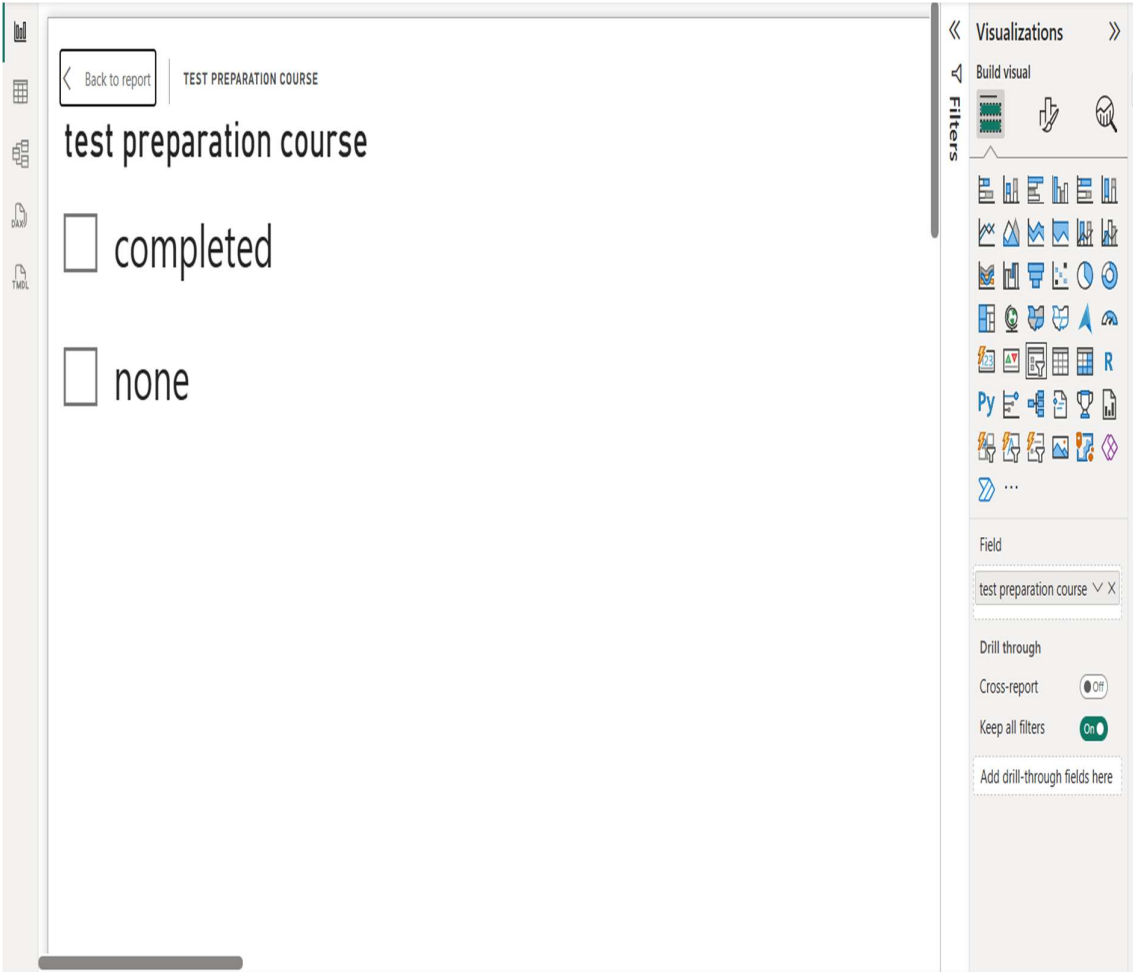


Figure 17: Adding a Subject slicer for interactive subject-wise performance analysis.

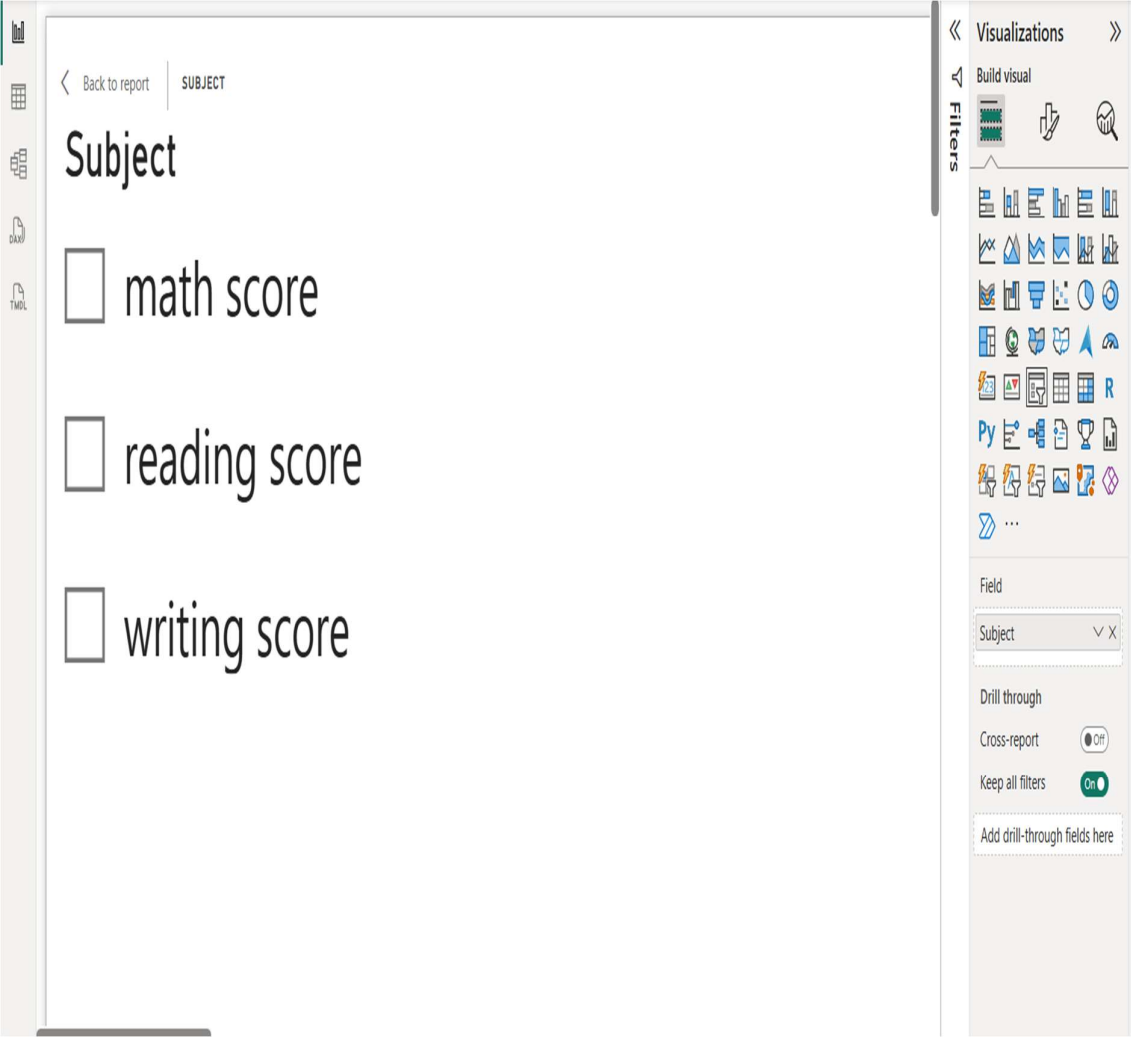


Figure 18: Creating the Academic Profile page for detailed student performance analysis.

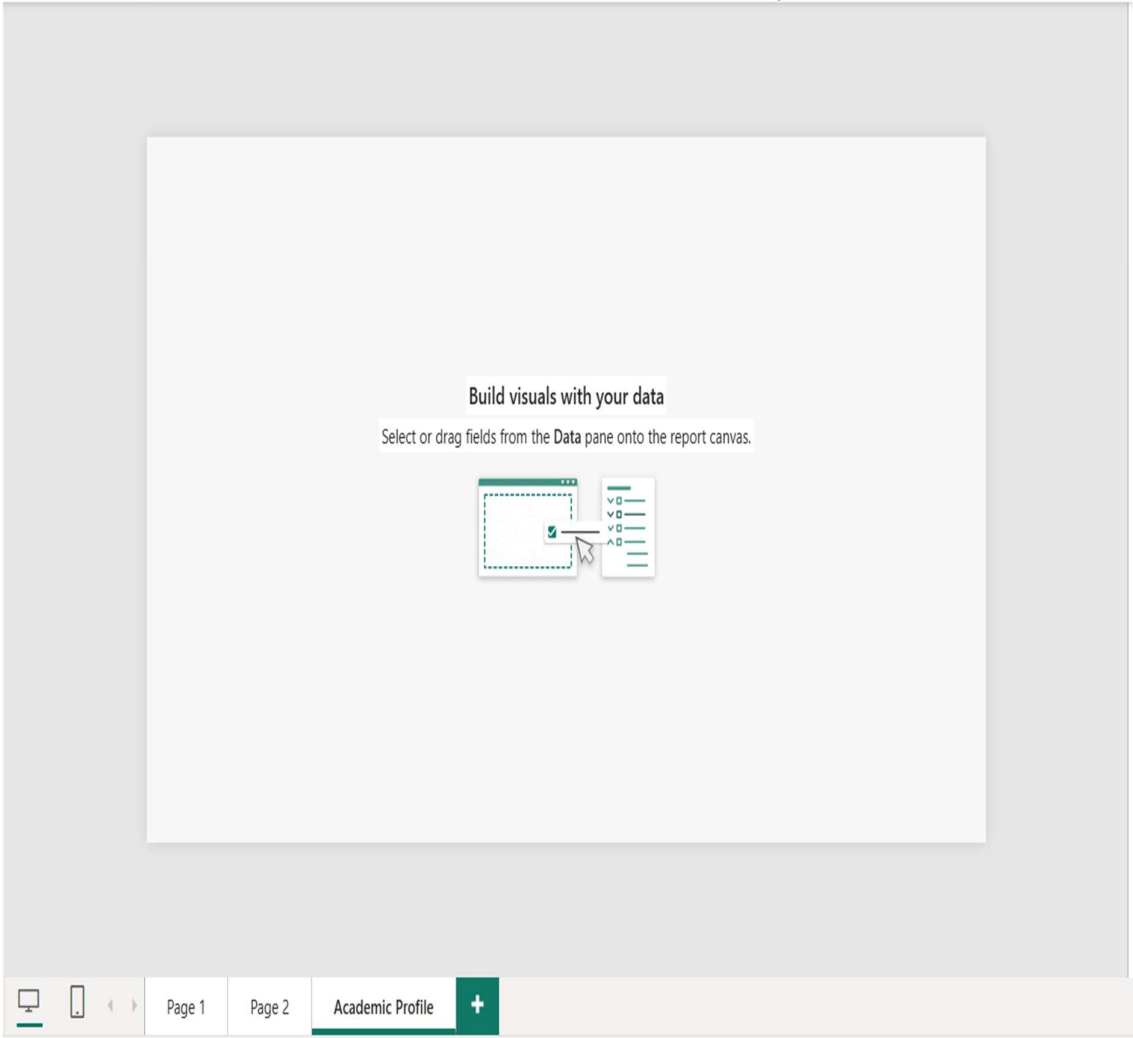


Figure 19: Configuring Student ID as the drill-through field for individual academic profiles.

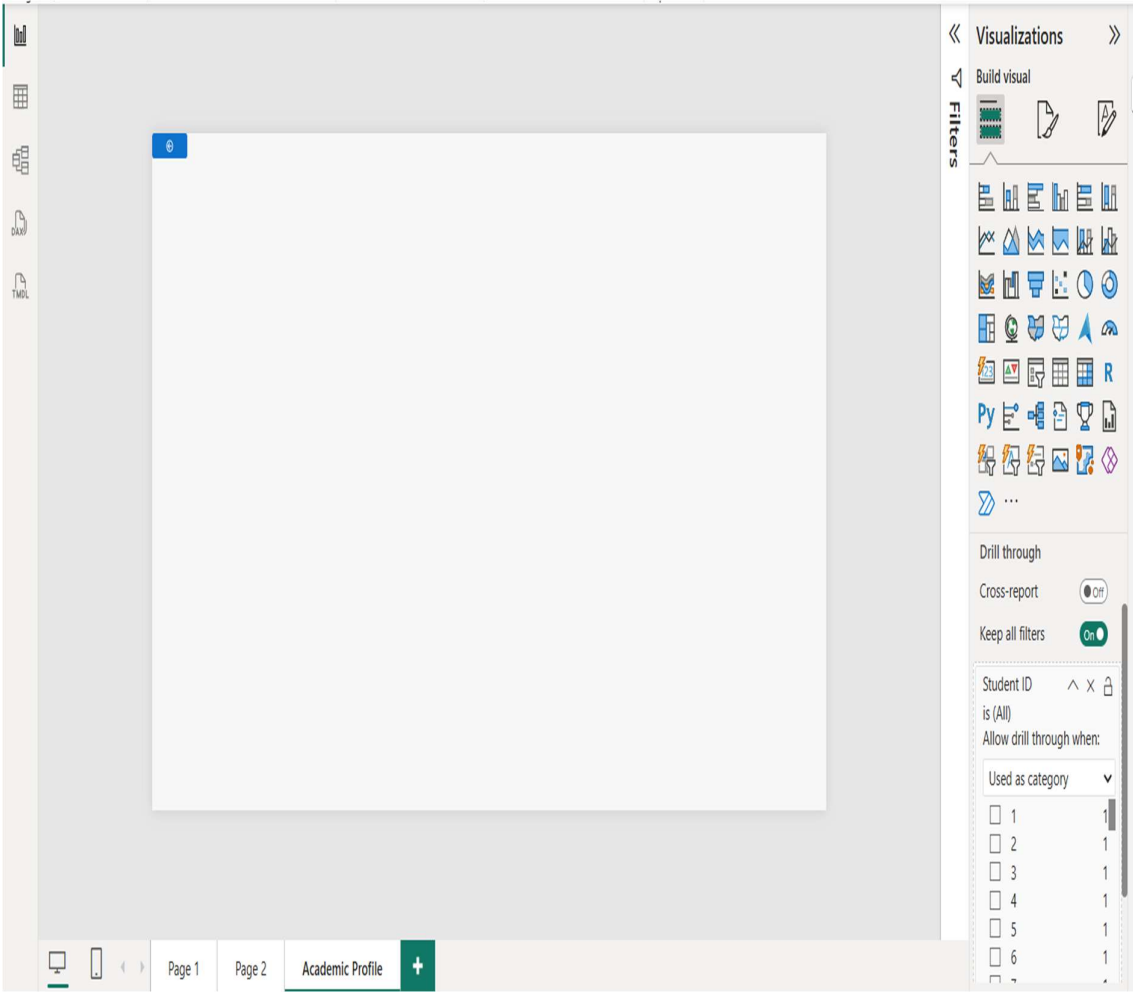


Figure 20: Academic Profile drill-through page displaying subject performance, average score, performance grade, pass/fail result, test preparation status, and navigation back to the main dashboard.

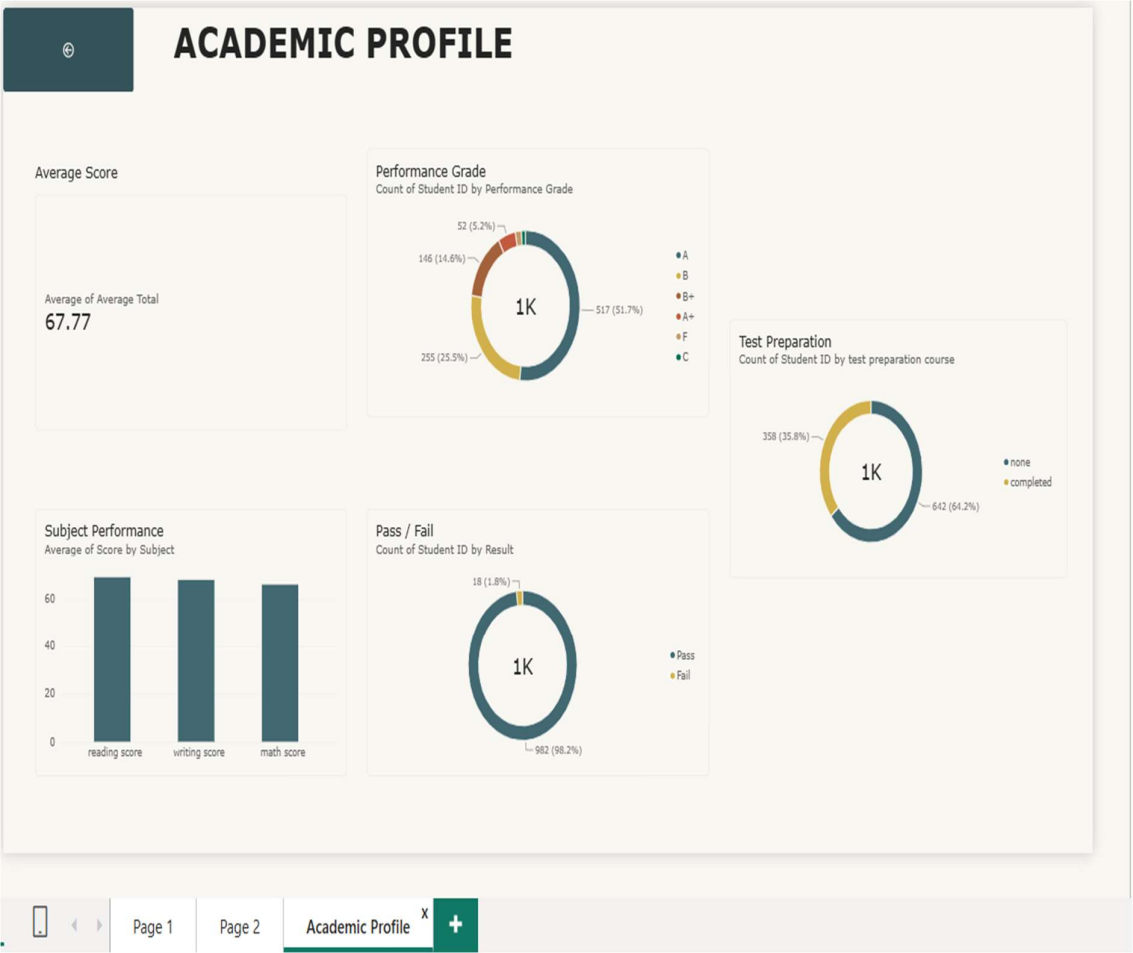


Figure 21: Adding a Student Performance table to the main dashboard to enable drill-through navigation to the Academic Profile page.

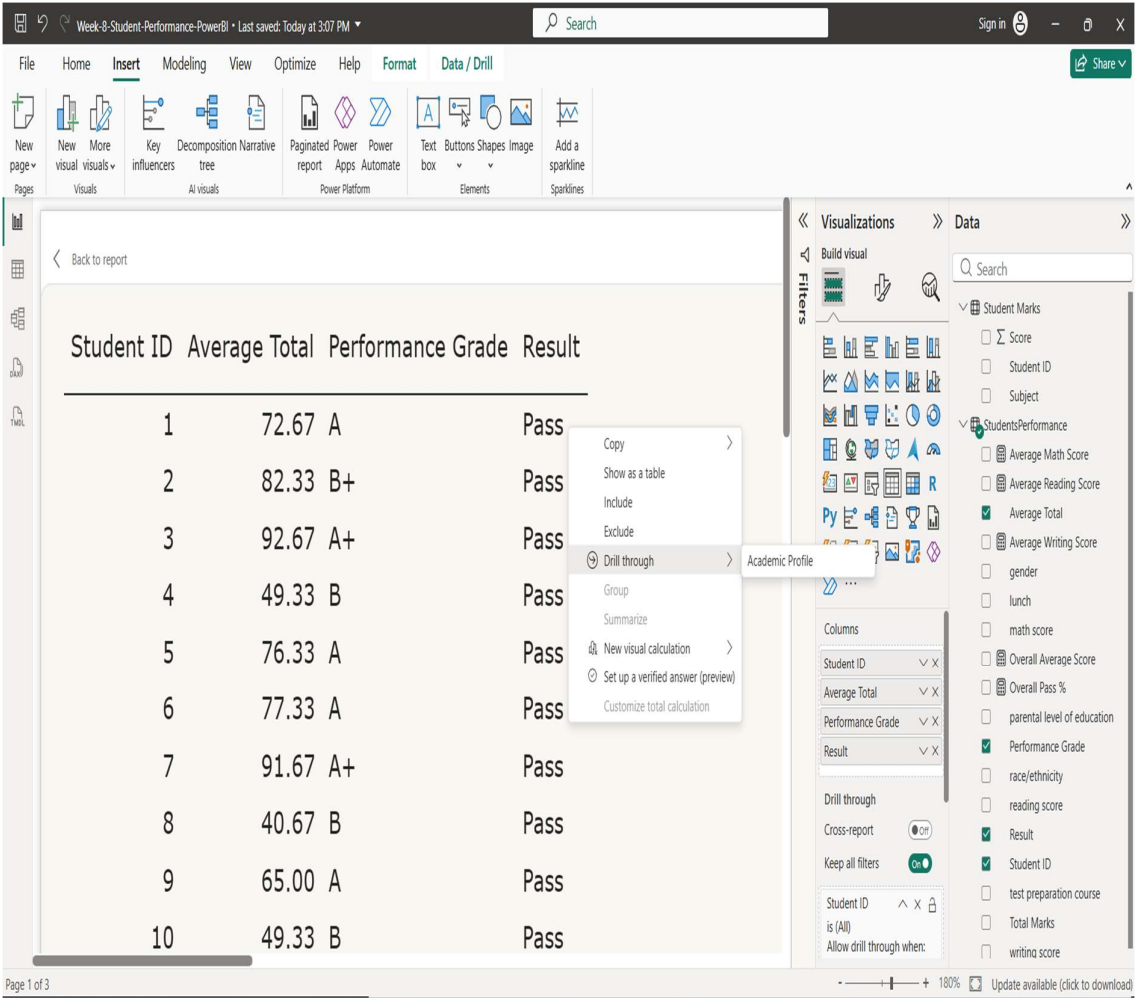
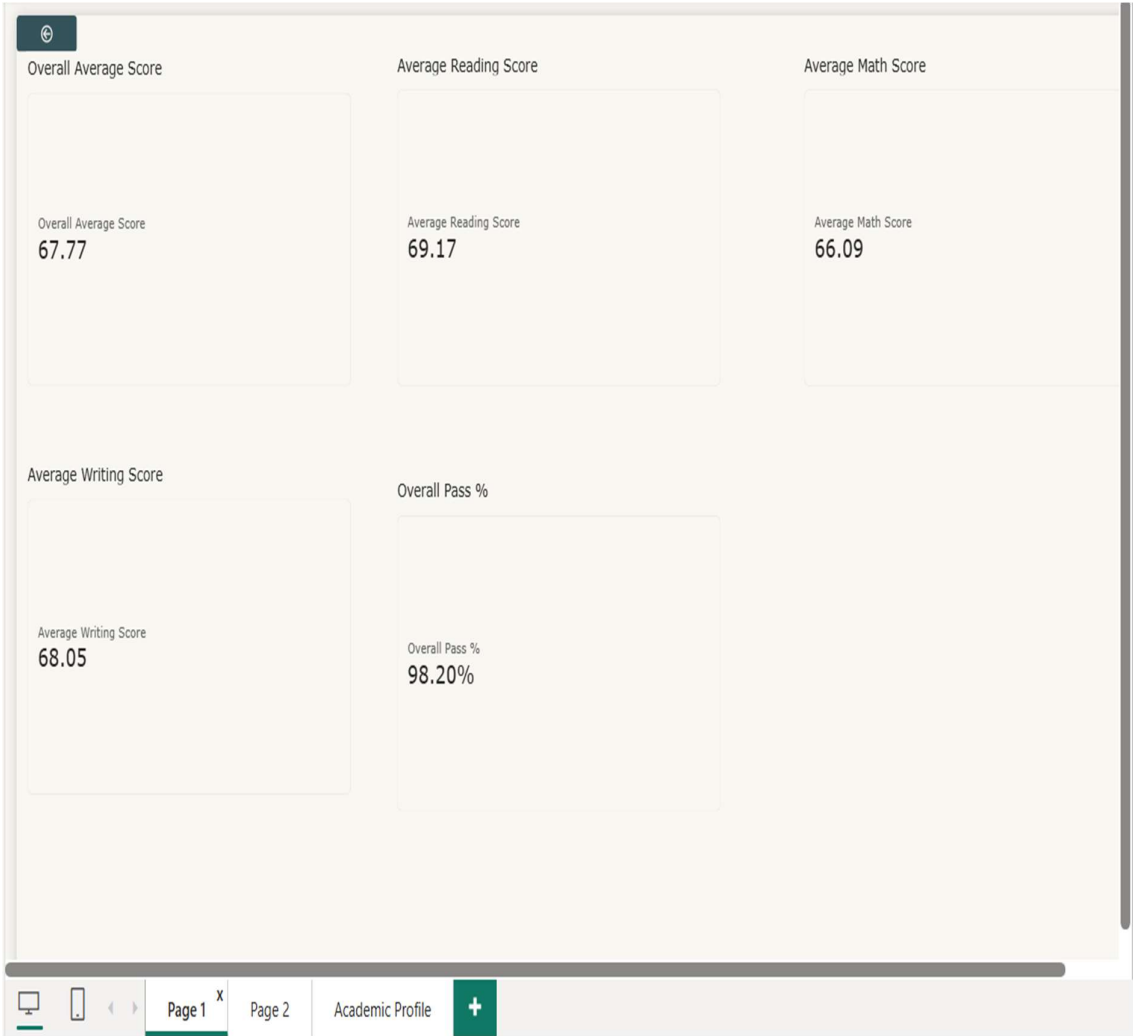
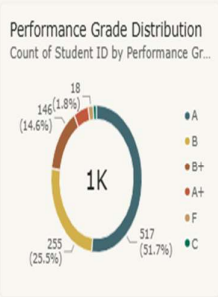
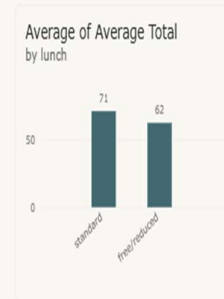
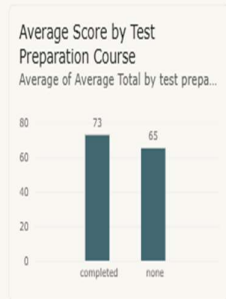
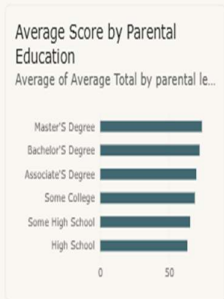


Figure 22: Final Student Performance Analytics dashboards displaying key performance indicators, subject-wise analysis, demographic insights, interactive slicers, and drill-through academic profiles.





Gender
gender
☐ female
☐ male

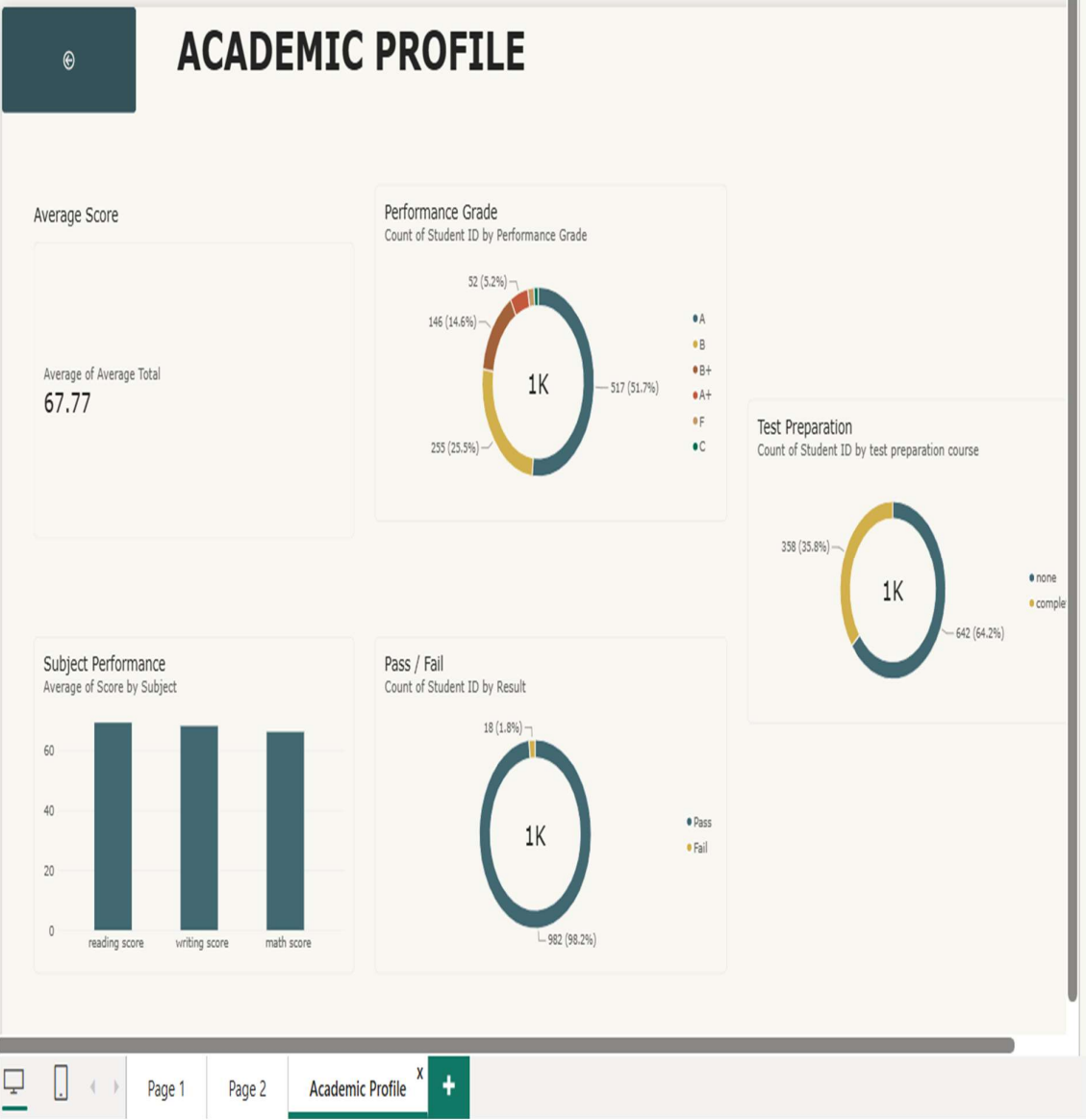
Race/Ethnicity
race/ethnicity
☐ group A
☐ group B
☐ group C
☐ group D
☐ group E

Parental Education
parental level of education
☐ Associate'S Degree
☐ Bachelor'S Degree
☐ High School
☐ Master'S Degree
☐ Some College
☐ Some High School

Lunch Type
lunch
☐ free/reduced
☐ standard

Test Preparation Course
test preparation course
☐ completed
☐ none

Subject
Subject
☐ math score
☐ reading score
☐ writing score



Conclusion

The Week 8 activity successfully transformed the prepared Student Performance dataset into an interactive Power BI analytics dashboard. The dashboards present key performance indicators, subject-wise performance, parental education analysis, test preparation comparison, lunch type analysis, and performance grade distribution.

Interactive slicers were added for Gender, Race/Ethnicity, Parental Education, Lunch Type, Test Preparation Course, and Subject, allowing users to explore the data from different perspectives. A Student Performance table and Academic Profile drill-through page were also created to provide detailed student-level analysis.

Overall, the Week 8 activity demonstrated how Power BI can be used to create interactive visualizations, analyze academic performance, identify meaningful patterns, and present data-driven insights in a clear and user-friendly dashboard.